

Morphophonology

Class 2: Boundary effects without boundaries



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Summary of last time

- Observation: phonology may apply unexpectedly in morphologically complex words
- Hypothesis: morphological boundaries can be mirrored by prosodic boundaries
- ALIGN constraints favor prosodic boundaries at morpheme boundaries
- May cause phonology to apply differently

Morpheme boundaries as prosodic boundaries

- Prosodic approach directly encodes the effect of morpheme boundaries by making phonology sensitive to boundaries
- ¿But why should prosody care so much about morpheme boundaries?
- ¿And why should phonological processes care so much about prosodic boundaries?
- ¿Are prosodic boundaries a *cause* or a *consequence* of how phonology applies to morphologically complex forms?



Prosodic boundaries: a "cheap fix"?

- German [aʊfʔɛsən]
 - [aʊf.ʔɛ.sən]: morpheme and syllable boundaries do not actually(?) align
- Granada Spanish [dehetʃo] *deshecho* 'undone'
 - [de.he.tʃo]: morpheme and syllable boundaries do not actually align



An alternative

- Morphological structure influences phonology indirectly
- Various possible mechanisms
 - Morphology restricts the domains that phonology evaluates
 - Morphology establishes correspondence relations between expressions that share morphological structure



Three approaches to compare

- Aligned prosodic and morpheme boundaries
 - Prosodic domains (last time)
- Cyclic evaluation (with input-output faithfulness)
- Transderivational correspondence (with output-output faithfulness)



Cyclic Evaluation

- Interleaving of morphology and phonology
- Expressions are evaluated when additional morphology is added
 - Numerous possible versions have been explored
- Important consequence: some phonological evaluations take place before the entire expression has been composed



Korean cluster simplification

- Korean phonotactics ban final CC#
 - [tʃip] 'house', [sok] 'inside', [nun] 'eye', [mal] 'speech'
 - *[palk], *[naps], *[talm], etc.
- Cluster simplification

/kaps/	kap	'price'
/hɪlk/	hɪk	'soil'
/talk/	tak	'chicken'
/salm/	sam	'life'
- Medial clusters are allowed
 - [əlɡul] 'face'



Cluster simplification in OT

- "Normal application": simplify in final position (only)

/talk/	*CC]	Max(C)	Dep(C)
a.  tak		*	
b. talk	*!		

/əlku/	*CC]	Max(C)	Dep(C)
a.  əlgul			
b. əgul		*	

Overapplication of cluster simplification

Contemporary Korean (Ko 2006, and many others)

/kaps-i/	kabi	*kapsi
/hɨlk-i/	hɨgi	*hɨlgi
/talk-i/	tagi	*talgi
/salm-i/	sami	*salmi

- The problem: why don't the medial clusters surface?
 - [talgi] would be perfectly legal (cf. [əlgul] 'face')

Cyclic evaluation of complex words

- Interleaving of morphology and phonology

Cycle 1	/talk/
Cluster simplification	tak
Cycle 2	/tak-i/
Cluster simplification	—
Output	tagi

- Cluster is simplified before context that could preserve it is added (counterbleeding of phonology by morphology)



Cyclic evaluation: in OT

- First cycle

/talk/	*CC]	Max(C)	Dep(C)
a.  tak		*	
b. talk	*!		

- Second cycle

/tak-i/	*CC]	Max(C)	Dep(C)
a.  taki			
b. talki			*!



Laryngeal neutralization

- Korean has a three-way laryngeal contrast

	Lax	Aspirated	Tense
p	발 pal 'foot'	팔 p ^h al 'arm'	빨래 p [*] al:ε 'laundry'
t	대 tε 'stalk'	테 t ^h e 'frame'	때 t [*] ε 'time'
k	개 kε 'dog'	캐 k ^h ε 'dig!'	깨 k [*] ε 'perilla'

- Only lax stops in final position

Lax	Aspirated	Tense
/sok/ soᵏ 'inside'	/puək ^h / puəᵏ 'kitchen'	/pak [*] / baᵏ 'outside'
/tʃip/ tʃiᵑ 'house'	/soᵑ ^h / soᵑ ^h 'pot'	

Laryngeal neutralization: in OT

- Contrast in initial position

/te/	*C ^h], C*]	Ident(c.g.)	Ident(s.g.)
a.  te			
b. t ^h e			*

/t ^h e/	*C ^h], C*]	Ident(c.g.)	Ident(s.g.)
a. te			*!
b.  t ^h e			

/t*e/	*C ^h], C*]	Ident(c.g.)	Ident(s.g.)
a. te		*!	
b.  t*e			

Laryngeal neutralization: in OT

- Neutralized in final position (hypothetical morphemes)

/tak/	*C ^h], C*]	Ident(c.g.)	Ident(s.g.)
a.  tak			
b. tak ^h	*!		*

/tak ^h /	*C ^h], C*]	Ident(c.g.)	Ident(s.g.)
a.  tak			*
b. tak ^h	*!		

/tak*/	*C ^h], C*]	Ident(c.g.)	Ident(s.g.)
a.  tak		*	
b. tak*	*!		



Laryngeal neutralization: in OT

- Intervocalic position: contrast, but with voicing

/taka/	*D ^h , *D*	*C ^h , C*	Ident(c.g.)	Ident(s.g.)	*s.g.	*V[-voi]V	Ident(voi)
a. taka						*!	
b. tak ^h a				*!	*	*	
c. tak*a			*!			*	
d.  taga							*

No voicing of aspirated, tense stops

/tak ^h a/	*D ^h , *D*	*Ch], C*]	Ident(c.g.)	Ident(s.g.)	*s.g.	*V[-voi]V	Ident(voi)
a. taka				*!		*	
b.  tak ^h a					*	*	
c. tag ^h a	*!				*		*
d. taga				*!			*

/tak*a/	*D ^h , *D*	*Ch], C*]	Ident(c.g.)	Ident(s.g.)	*c.g.	*V[-voi]V	Ident(voi)
a. taka			*!			*	
b.  tak*a					*	*	
c. tag*a	*!						*
d. taga			*!				*



Overapplication of laryngeal neutralization

Contemporary Korean (Ko 2006, and many others)

/sup ^h -i/	subi	'forest-NOM'
/cip ^h -il/	cibil	'straw-ACC'
/mulip ^h -e/	muribe	'knee-LOC'
/puək ^h -in/	puəgin	kitchen-TOP'

- The problem: why don't underlyingly aspirated C^h surface faithfully before suffix?
- [sup^{hi}] would be phonotactically fine



Overapplication of laryngeal neutralization

● Cycle 1

/sup ^h /	*D ^h , *D*	*C ^h], C*]	Ident(c.g.)	Ident(s.g.)	*s.g.	*V[-voi]V	Ident(voi)
a.  sup				*			
b. sup ^h		*!			*		

● Cycle 2

/sup-i/	*D ^h , *D*	*C ^h], C*]	Ident(c.g.)	Ident(s.g.)	*s.g.	*V[-voi]V	Ident(voi)
a.  subi							*
b. sup ⁱ						*!	
c. sup ^{hi}				*!	*	*	
d. sub ^{hi}	*!			*	*		*



Summary of Korean overapplication

- Neutralization process applies in first cycle
- Neutralized output is carried forward into cycle 2
- Result: overapplication of neutralization in suffixed form



English aspiration

- Recall misapplication of aspiration after prefixes

dis[p]lay

dis[p^h]ease

des[p]otism

dis[p^h]ossess

whis[p]ering

mis[p^h]erceive

pis[t]achio

mis[t^h]abulate

Pis[k]ataway

mis[k^h]ategorize

- Cyclic approach: stage of evaluation in which stem-initial stops are word-initial

Cyclic misapplication of aspiration?

- Cycle 1

/pli:z/	*sT ^h	T ^h /#_V	*C ^h	IO-Ident([±asp])
a.  ph ^h liz			*	*
b. pli:z		*!		

- Cycle 2: a problem!

/dis-phli:z/	*sT ^h	T ^h /#_V	*C ^h	IO-Ident([±asp])
a.  disp ^h liz	*!		*	
b.  disp ^h liz				*



The problem

- Cycle 1: predictable initial aspiration
 - /pli:z/ → [p^hli:z]
- Cycle 2: prefix changes the context, we predict de-aspiration
 - /dis-p^hli:z/ → *[displi:z]
- Diagnosis: because aspiration is allophonic in English, grammar both aspirates and deaspirates as needed
- How do we fix this?





The solution: re-ranking across cycles

- Cycle 1

/pli:z/	*sT ^h	T ^h /#_V	*C ^h	IO-Ident([±asp])
a.  phliz			*	*
b. pli:z		*!		

- Cycle 2: preserve aspiration with higher Ident

/dis-phli:z/	IO-Ident([±asp])	*sT ^h	T ^h /#_V	*C ^h
a.  disp ^h liz		*		*
b. displi:z	*!			

The logic of the approach

- Phonology assigns certain properties (feature values) at an earlier stage of the derivation, and those properties are then retained through later stages
- Assigning properties, in OT:
 - Markedness >> Faithfulness *for the relevant property*
- Retaining properties:
 - Faithfulness >> Markedness



Assigning properties in OT: $\mathcal{M} \gg \mathcal{F}$

Markedness constraints demand predictable changes to feature values, as required by context

/sa/	*V[+cont,-voi]V	*[+cont,+voi]	IO-IDENT([±voi])
a.  sa			
b. za		*!	*

/asa/	*V[+cont,-voi]V	*[+cont,+voi]	IO-IDENT([±voi])
a. asa	*!		
b.  aza		*	*



Assigning properties in OT: $\mathcal{M} \gg \mathcal{F}$

Predictably: even if input is specified wrongly for this feature, markedness demands the right output

/asa/	*V[+cont,-voi]V	*[+cont,+voi]	IO-IDENT([±voi])
a. asa	*!		
b.  aza		*	*

/aza/	*V[+cont,-voi]V	*[+cont,+voi]	IO-IDENT([±voi])
a. asa	*!		*
b.  aza		*	



When values are predictable...

/asa/	*V[+cont,-voi]V	*[+cont,+voi]	IO-IDENT([±voi])
a. asa	*!		
b.  aza		*	*

/aza/	*V[+cont,-voi]V	*[+cont,+voi]	IO-IDENT([±voi])
a. asa	*!		*
b.  aza		*	

Richness of the base (ROTB): analysis doesn't impose any requirements on UR's for this particular property, in this particular language



Retaining properties in OT: $\mathcal{F} \gg \mathcal{M}$

Faithfulness constraints demand that underlying feature values are preserved on the surface

/sa/	IO-IDENT([±voi])	*V[+cont,-voi]V	*[+cont,+voi]
a.  sa			
b. za	*!		*

/asa/	IO-IDENT([±voi])	*V[+cont,-voi]V	*[+cont,+voi]
a.  asa		*	
b. aza	*!		*



Retaining properties in OT: $\mathcal{F} \gg \mathcal{M}$

Faithfulness constraints demand that underlying feature values are preserved on the surface

/za/	IO-IDENT([±voi])	*V[+cont,-voi]V	*[+cont,+voi]
a. sa	*!		
b.  za			*

/aza/	IO-IDENT([±voi])	*V[+cont,-voi]V	*[+cont,+voi]
a. asa	*!	*	
b.  aza			*



Restating the logic of the approach, in OT

- Properties are assigned at earlier stages of derivation
 - Markedness $\gg$ Faithfulness
- Those properties are retained at later stages
 - Faithfulness $\gg$ Markedness (*reranking*)
- "Stratal" OT (Kiparsky, Bermúdez-Otero, and others)

/pli:z/	*sTh	Th/#_V	*Ch	IO-Ident([±asp])
a.  phliz			*	*
b. pli:z		*!		

/dis-phli:z/	IO-Ident([±asp])	*sTh	Th/#_V	*Ch
a.  disp ^h liz		*		*
b. disp ^h li:z	*!			



A cloud on the horizon: ROTB

- Re-ranking allows for contrast at later cycles
- Affixes introduced at later cycles are not subject to predictable allophonic repairs (Benua 1996)

/pli:z/	*sTh	Th/#_V	*Ch	IO-Ident([±asp])
a.  phliz			*	*
b. pli:z		*!		

/pɹi:-phli:z/	IO-Ident([±asp])	*sTh	Th/#_V	*Ch
a.  pɹi:phliz		*		*
b.  pʰɹi:pli:z	*!			

(We'll consider some possible patches later)



Summary of cyclic derivation

- Phonological evaluations apply throughout the morphological derivation of words
 - "Misapplication" in complex words when there is an inner cycle
- Features assigned during earlier cycles may be preserved in subsequent cycles
 - Or, affect how other processes apply (opacity)
- A tension: preserving properties of stems, while allowing phonological processes to continue to apply to affixes



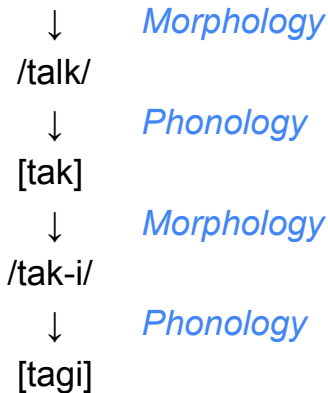
Transderivational correspondence

- A shared insight: cluster simplification in Korean /talk-i/ → [tagi] is due to the fact that it contains a subpart [tak]
- Difference: how that phonological evaluation references that subpart
- Cyclic faithfulness: output of earlier stage is input to the next stage
 - IO-Faithfulness to earlier stages of derivation
- Transderivational faithfulness: base is a separate word
 - OO-Faithfulness to related derivations

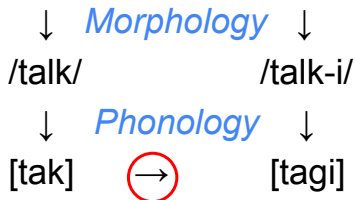


Cluster simplification in *tagi*

Cyclic approach



Transderivational approach



IO vs. OO Faithfulness

- Simple and complex expressions are each evaluated separately by the phonology
- Surface form of complex expressions constrained to resemble...
 - Their UR
 - Surface forms of related expressions (usually: base of affixation: Kenstowicz 1996, Benua 1996, Burzio 1996)
- Similarity is enforced by correspondence constraints (faithfulness; McCarthy and Prince 1995)



Overapplication of CC simplification with BD-Ident

- Ko (2006)

/talk/	BD-Dep(C)	*CC]	IO-Max(C)	IO-Dep(C)
a.  tak			*	
b. talk		*!		

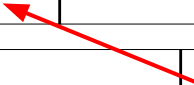
/talk-i/ Base: [tak]	BD-Dep(C)	*CC]	IO-Max(C)	IO-Dep(C)
a.  tagi			*	
b. talgi	*!			



English: *dis[p]lay* vs. *dis[p^h]ease*

/dɪspleɪ/	BD-Ident ([±asp])	*sC ^h	T ^h Ũ	IO-Ident ([±asp])
a.  dɪ'spleɪ			*	
b. dɪ'sp ^h leɪ		*!		*

/plɪz/	BD-Ident ([±asp])	*sC ^h	T ^h Ũ	IO-Ident ([±asp])
a. plɪz			*!	
b.  p ^h lɪz				*



/dɪs-plɪz/	BD-Ident ([±asp])	*sC ^h	T ^h Ũ	IO-Ident ([±asp])
a. dɪsplɪz	*!		*	
b.  dɪsp ^h lɪz		*		*



Aspiration in prefixes

/pliz/	BD-Ident([±asp])	*sC ^h	T ^h Ŵ, #T ^h V	IO-Ident([±asp])
a. pliz			*!	
b.  phliz				*

/p _ɪ i-pliz/	BD-Ident([±asp])	*sC ^h	T ^h Ŵ, #T ^h V	IO-Ident([±asp])
a. p _ɪ ipliz	*!		**	
b. p _ɪ iphliz			*!	*
c.  p ^h _ɪ iphliz				**

/p ^h _ɪ i-pliz/	BD-Ident([±asp])	*sC ^h	T ^h Ŵ, #T ^h V	IO-Ident([±asp])
a. p _ɪ ipliz	*!		**	*
b. p _ɪ iphliz			*!	**
c.  p ^h _ɪ iphliz				*

A cloud on the horizon

- Misapplication in [dɪsp^hli:z] relies on the existence of an independent word form [p^hli:z]
- Prediction: normal application with bound stems
- Potential problems...

<i>expression</i>	ɛkspɹɛʃn	<i>expropriate</i>	ɛksp ^h ɹɔʊpɹiɛt
<i>expunge</i>	ɛkspɹʌndʒ	<i>discordant</i>	dɪsk ^h ɔɹdʌnt (?)
<i>extenuating</i>	ɛkstɛnjueɪɪŋ		
<i>dispensary</i>	dɪspɛnsəɹi		



Summary of transderivational faithfulness

- Regular application: ranking of Markedness and IO-Faithfulness
- Misapplication in complex forms may occur when there is a free-standing base form
 - Higher-ranked BD-Faithfulness
- Regular phonology still applies when it does not violate BD-Ident
 - e.g., in affixes



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